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Midterm
Blueprint

PlateVision

AI-Assisted Vehicle & License Plate Analysis from Security Footage

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Motivation: A Real-World Problem

From a real incident to a CV question

Police contacted a family while looking for information about a black Corvette seen on their street. Ring footage captured the vehicle, but finding a clear view of the plate across many frames was difficult. This inspired our question: Can computer vision help identify and organize the clearest available vehicle and plate evidence from security footage?

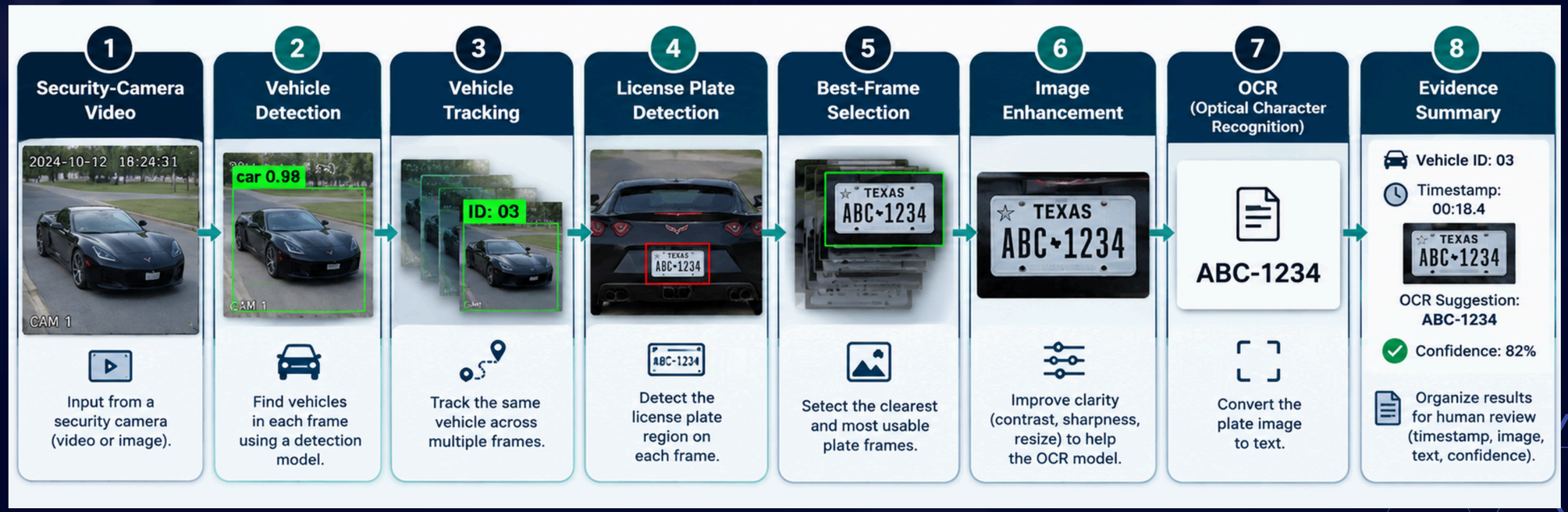


- Who cares: people reviewing residential or business security footage, including homeowners, security personnel and investigators.

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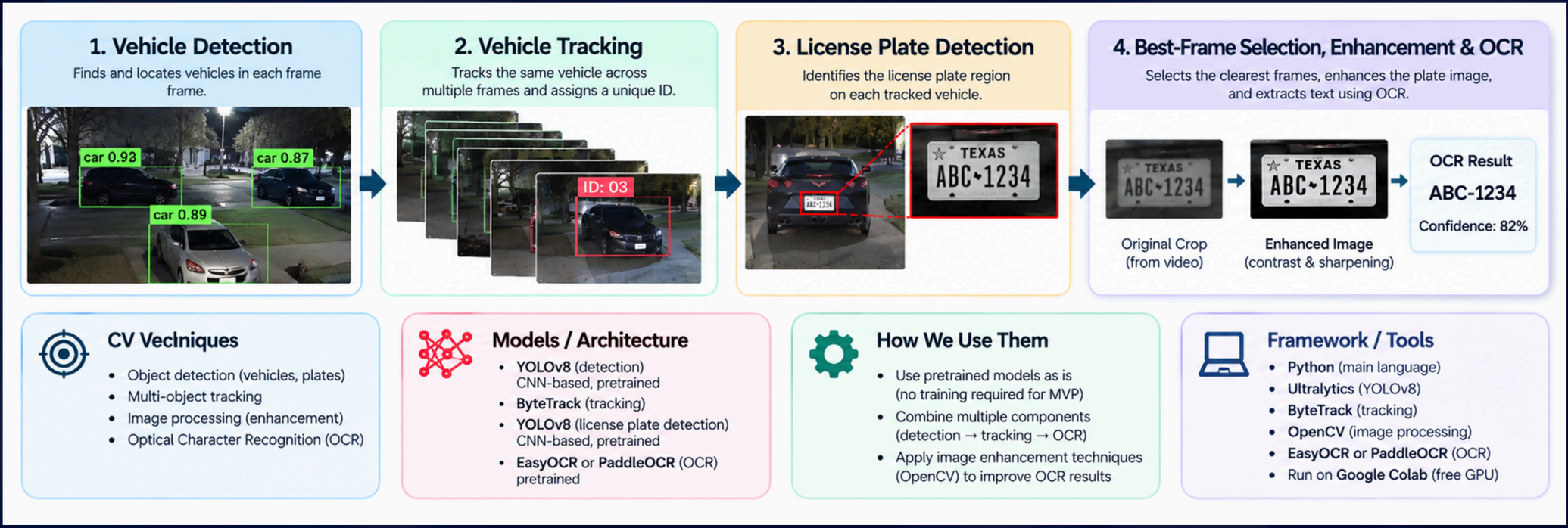
From Problem to Proposed Solution

PlateVision analyzes security-camera video to locate vehicles and license plates, select the clearest available plate frames, and organize the results for human review.



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How PlateVision Works



Vehicle: pretrained YOLOv8 (COCO) — detects cars without initial training.
Plates: YOLOv8 with plate-specific weights — localizes plates (weights to be tested).
OCR: EasyOCR — suggests characters for human review.

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Data Strategy

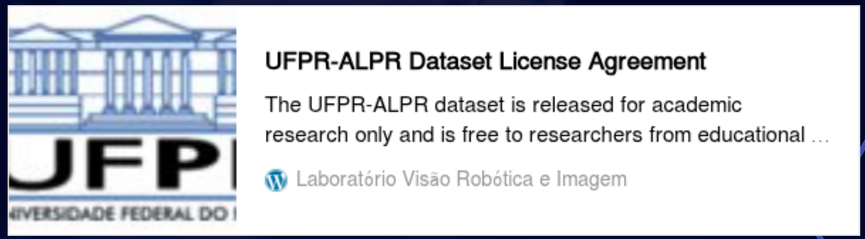
Primary Dataset — UFPR-ALPR

- 4,500 annotated images
- 150 vehicles
- 1920 × 1080 resolution
- Real-world vehicle scenarios
- Existing vehicle, license-plate, and character annotations



Backup Dataset — Kaggle

Backup: Kaggle License Plate Detection Dataset (10,125 images)
Downloaded locally; YOLO-format plate boxes
Supports plate detection if UFPR access is delayed
OCR evaluation requires separate character labels



Access requested

Real-World Testing

Security-camera video samples for realistic testing
First demo: video in → plate bounding box out
Tracking and OCR added after detection works
Identifiable footage will remain private

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Success Metrics

🎯 LICENSE PLATE DETECTION

≥ 0.70 mAP@50

Correctly detects and localizes license plates

🔤 OCR ACCURACY

≥ 70%

Correctly reads complete visible plates

🎞️ USABLE FRAME

≥ 80%

Detected vehicles produce a usable plate frame

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Milestone Plan



WEEK 10

1. BLUEPRINT

Finalize project plan, dataset, workflow & metrics.

WEEK 11

2. FIRST WORKING DEMO

Video in → plate box out. Validate basic detection first.

WEEKS 12-13

3. MAKE IT YOURS

Add vehicle detection, tracking, best-frame selection, enhancement & EasyOCR.

WEEK 14

4. IMPROVE & MEASURE

Test the system, calculate metrics & document failure cases.

WEEK 15

5. PACKAGE & PRESENT

Complete demo, README, final slides & presentation.

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Risks & Resources

⚠️ RISK 1 — Poor Video Quality

Plates may be too small, blurry, dark, or partially visible.

Plan B
Focus on vehicle detection, tracking, and best-frame selection. Flag unreadable plates instead of forcing an OCR result.

⚠️ RISK 2 — Inconsistent OCR Results

OCR may produce different plate readings across video frames.

Plan B
Compare OCR results from multiple high-quality frames and present the best candidate with confidence information for human verification.

💻 Resources

Compute
Google Colab

Models/Tools
YOLO • ByteTrack • OpenCV • OCR

Dataset
UFPR-ALPR
access requested

Estimated Cost
$0

All project resources will use free/open-source or academic-use options.